



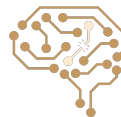
Bayes' Nets - Representation

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20 Ordibehesht 1405

CE 417-Artificial Intelligence
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Some of Slides have been adopted from Berkeley CS188 &
Sharif UT CE417



Joint Distributions

- A joint distribution over a set of random variables: $X_1, X_2, \dots, X_n$ specifies a real number for each assignment (or outcome):

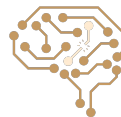
$$P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

$$P(x_1, x_2, \dots, x_n)$$

- Must obey:

$$P(x_1, x_2, \dots, x_n) \geq 0$$

$$\sum_{(x_1, x_2, \dots, x_n)} P(x_1, x_2, \dots, x_n) = 1$$

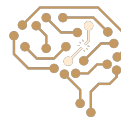


The Chain Rule

- Any joint distribution can be considered as an incremental product of conditional distributions!

$$P(x_1, x_2, x_3) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2)$$

$$P(x_1, x_2, \dots, x_n) = \prod_i P(x_i|x_1 \dots x_{i-1})$$

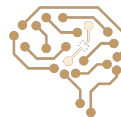


Independence

- Two variables are independent if:

$$\forall x, y : P(x, y) = P(x)P(y)$$

- This says that their joint distribution factors into a product two simpler distributions
- Another form: $\forall x, y : P(x|y) = P(x)$
- We write: $X \perp\!\!\!\perp Y$



Conditional Independence

- X is conditionally independent of Y given Z

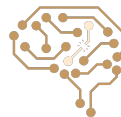
$$X \perp\!\!\!\perp Y | Z$$

- If and only if:

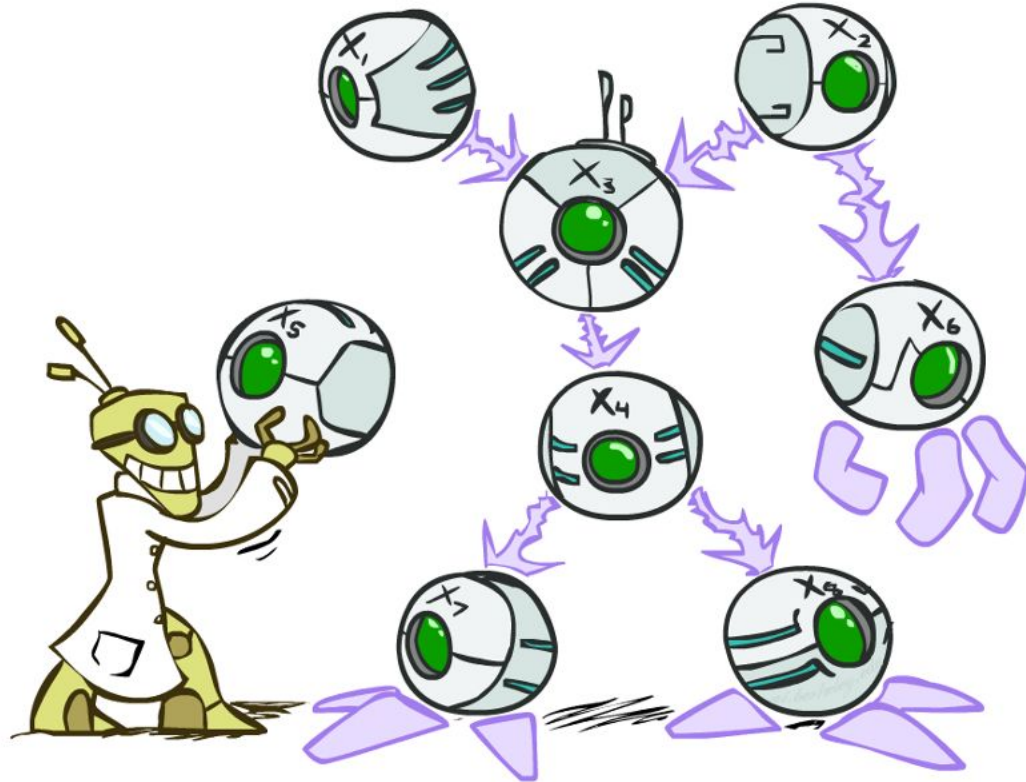
$$\forall x, y, z : P(x, y | z) = P(x | z) P(y | z)$$

- Or, equivalently, if and only if:

$$\forall x, y, z : P(x | z, y) = P(x | z)$$



Today: Bayes Nets



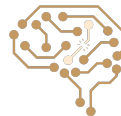
How many rows if all random variables are binary?

$$P(X_1, X_2, \dots, X_n)$$

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$O(2^n)$$



How many rows if all random variables are binary? (cont.)

- Chain Rule:

$$P(X_1, X_2, \dots, X_n)$$

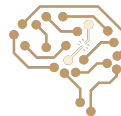
$$P(\text{Traffic}, \text{Rain}, \text{Umbrella}) =$$

$$P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain}, \text{Traffic})$$

- By using conditional Independence:

$$P(\text{Traffic}, \text{Rain}, \text{Umbrella}) =$$

$$P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain})$$



How many rows if all random variables are binary? (cont.)

- Chain Rule:

$$2^n - 1$$

$$P(X_1, X_2, \dots, X_n)$$

$$P(\text{Traffic}, \text{Rain}, \text{Umbrella}) =$$

$$P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain}, \text{Traffic})$$

- By using conditional Independence:

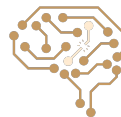
$$P(\text{Traffic}, \text{Rain}, \text{Umbrella}) =$$

$$P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain})$$

1

2

2



Conditional Independence

Domain:

Traffic

Umbrella

Raining



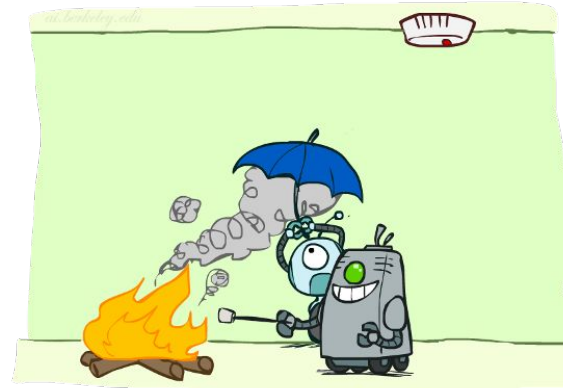
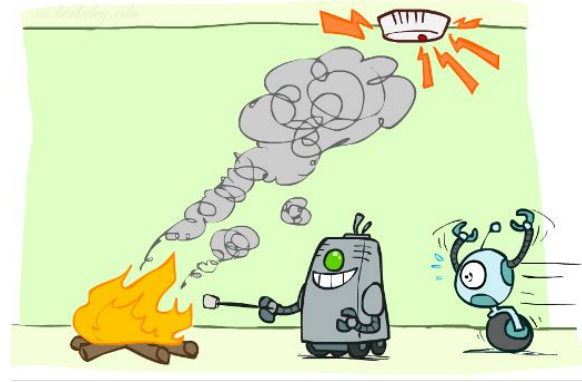
Conditional Independence

Domain:

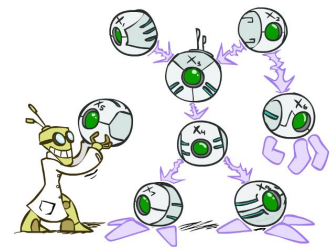
Fire

Smoke

Alarm



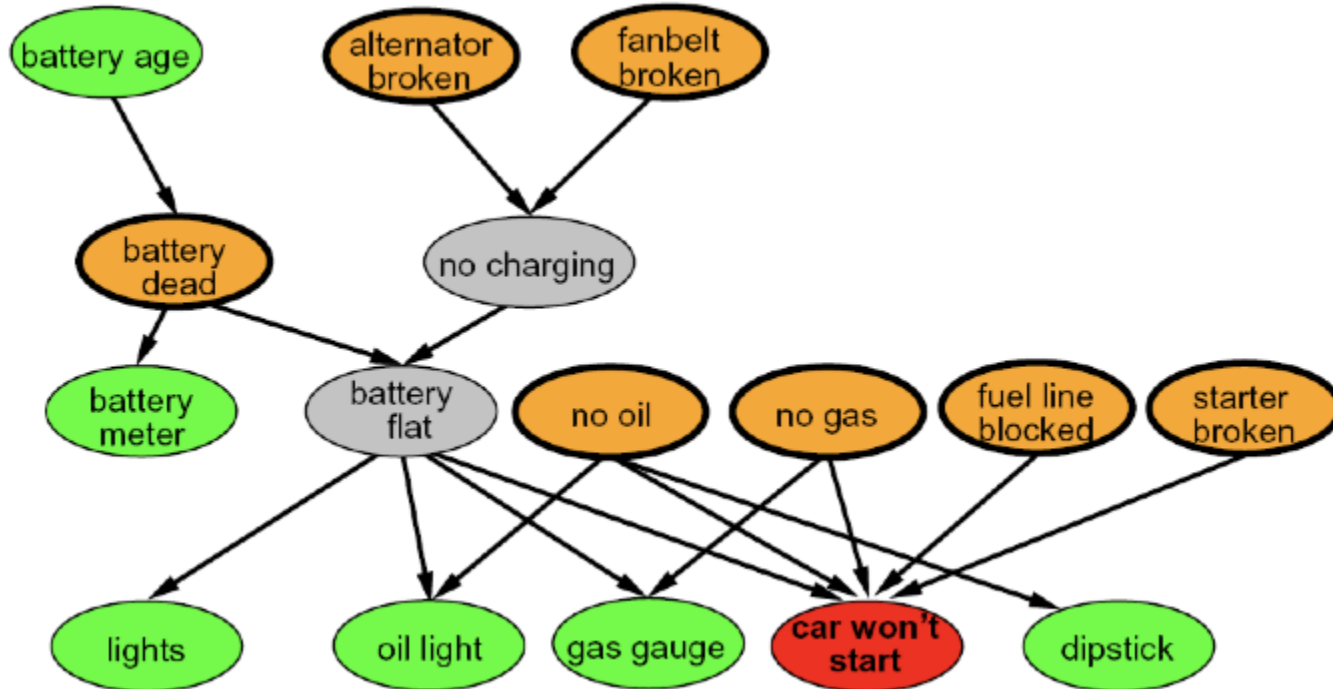
Bayes' Nets: Big Picture



- Using full joint distribution tables as our probabilistic models:
 - The joint is WAY too big to represent explicitly or learn (estimate) anything empirically, unless there are only a few variables.
- **Bayes' nets:** a technique for describing complex joint distributions (models) using simple, local distributions (conditional probabilities)
 - Also called graphical models
 - We describe how variables locally interact
 - Local interactions chain together to give global, indirect interactions

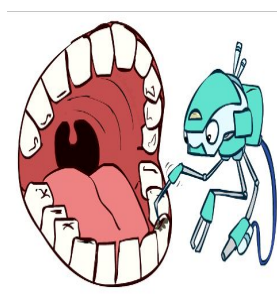
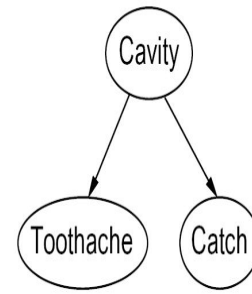


Example Bayes' Net: Car



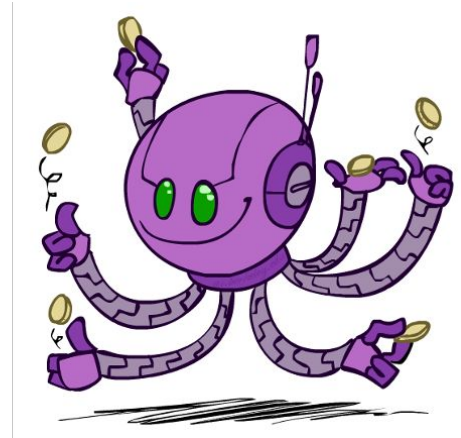
Graphical Model Notation

- Nodes: variables (with domains)
 - Can be assigned (observed) or unassigned (unobserved)
- Arcs: interactions
 - Similar to CSP constraints
 - Indicate “direct influence” between variables
 - Formally: encode conditional independence (more later)
- For now: imagine that arrows mean direct causation



Example: Coin Flips

N Independent Coin Flips



No Interaction Between Variables: Absolute Independence



Example: Traffic

- Variables:

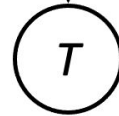
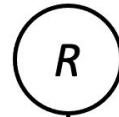
- R: It rains
- T: There is traffic



- Model 1: independence

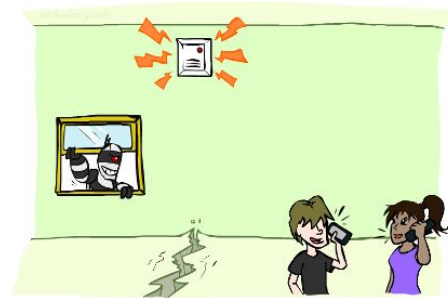


- Model 2: rain causes traffic

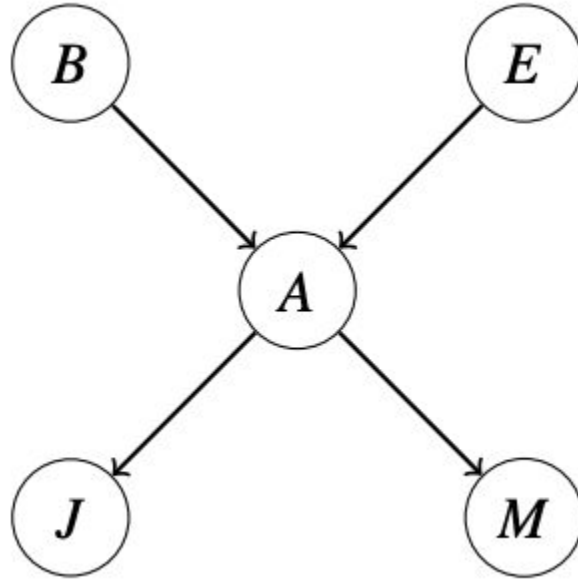


Example: Burglary

- A **burglar alarm**, respond occasionally to minor **earthquakes**.
- Neighbors **John** and **Mary** call you when hearing the alarm.
- John nearly always calls when hearing the alarm.
- Mary often misses the alarm.
- Random Variables: **Burglary, Earthquake, Alarm, JohnCalls, MaryCalls**

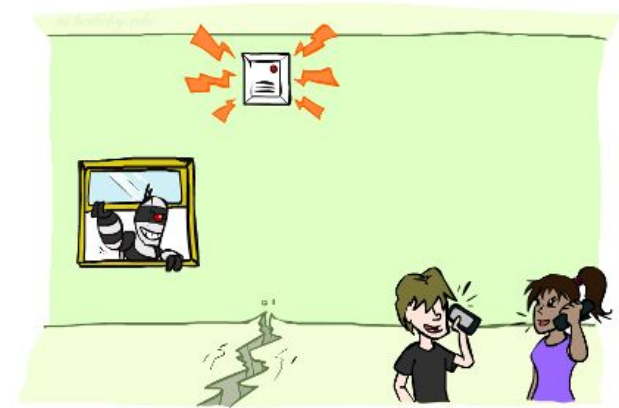


Burglary Modeling (Example)



A Reasoning Scenario

- You are at work, John calls to say that your alarm is ringing, but Marry doesn't call.
- Sometimes it sets off by minor earthquakes. Is there a burglar?
- Random Variables:
 - Burglary
 - Earthquake
 - Alarm
 - JohnCalls
 - MaryCalls



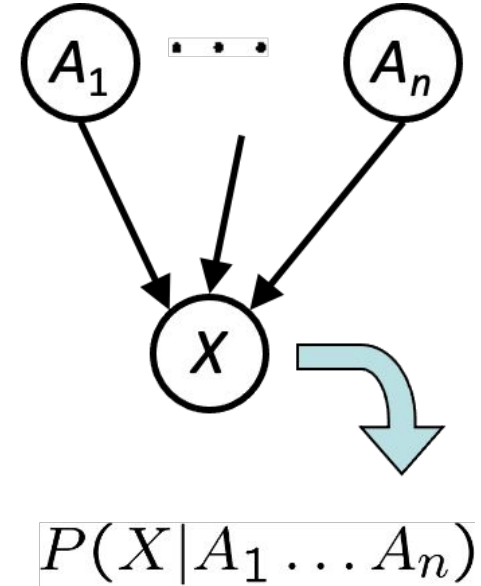
A Reasoning Scenario

$$P(\text{Burglary} | \text{JohnCalls} = T, \text{MarryCalls} = F)$$



Bayes' Net Semantics

- A set of nodes, one per variable X
- A directed, acyclic graph
- Each link from X to Y shows a "direct influence" of X on Y
- For each node, a conditional probability distribution $P(X_i | \text{Parents}(X_i))$ shows the effects of parents on the node
- CPT: conditional probability table



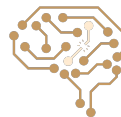
A Bayes net = Topology (graph) + Local Conditional Probabilities



Probabilities in BNs

- Bayes' nets implicitly encode joint distributions
- To see what probability a BN gives to a full assignment, multiply all the relevant conditionals together:

$$P(x_1, x_2, \dots x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$



Probabilities in BNs (cont.)

Why are we guaranteed that setting

$$P(x_1, x_2, \dots x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$

results in a proper joint distribution?

Chain rule (valid for all distributions):

$$P(x_1, x_2, \dots x_n) = \prod_{i=1}^n P(x_i | x_1 \dots x_{i-1})$$

Assume conditional independences:

$$P(x_i | x_1, \dots x_{i-1}) = P(x_i | \text{parents}(X_i))$$

→ Consequence:
$$P(x_1, x_2, \dots x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$

Not every BN can represent every joint distribution

- The topology enforces certain conditional independencies



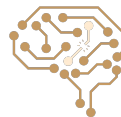
Probabilities in BNs (cont.)

- Bayes' nets implicitly encode joint distributions
- Chain Rule:

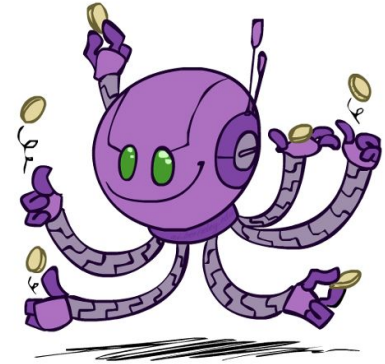
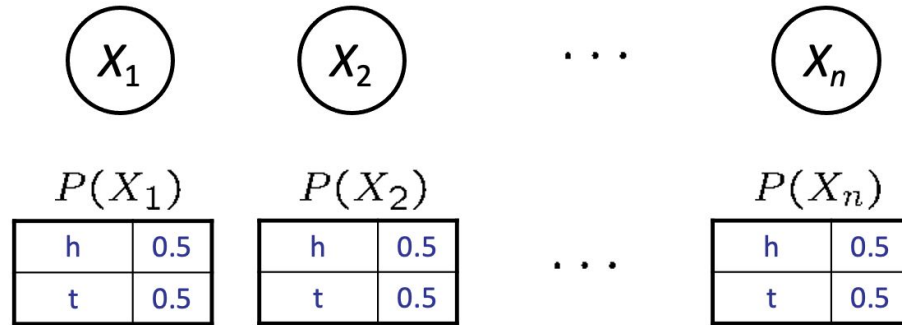
$$P(x_1, x_2, \dots x_n) = \prod_i P(x_i | x_1 \dots x_{i-1})$$

- Bayes Net Version:

$$P(x_1, x_2, \dots x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$



Example: Coin Flip

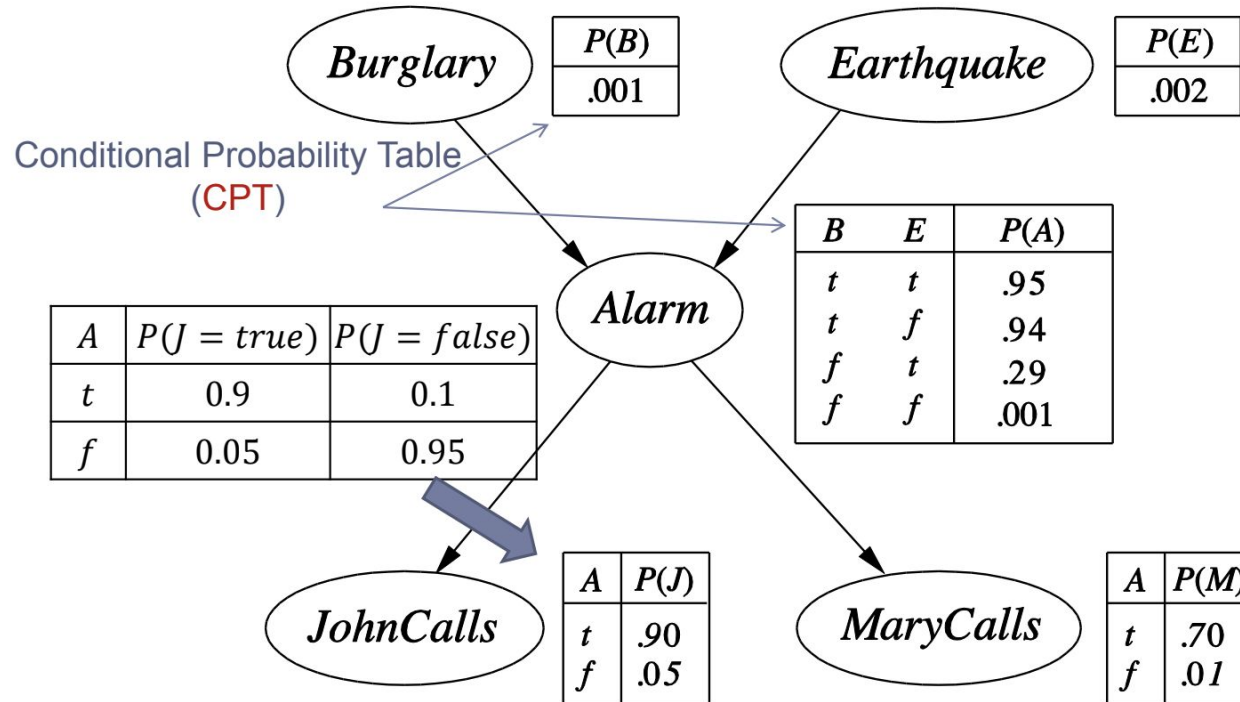


$$P(h, h, t, h) =$$

Only distributions whose variables are absolutely independent can be represented by a Bayes' net with no arcs.

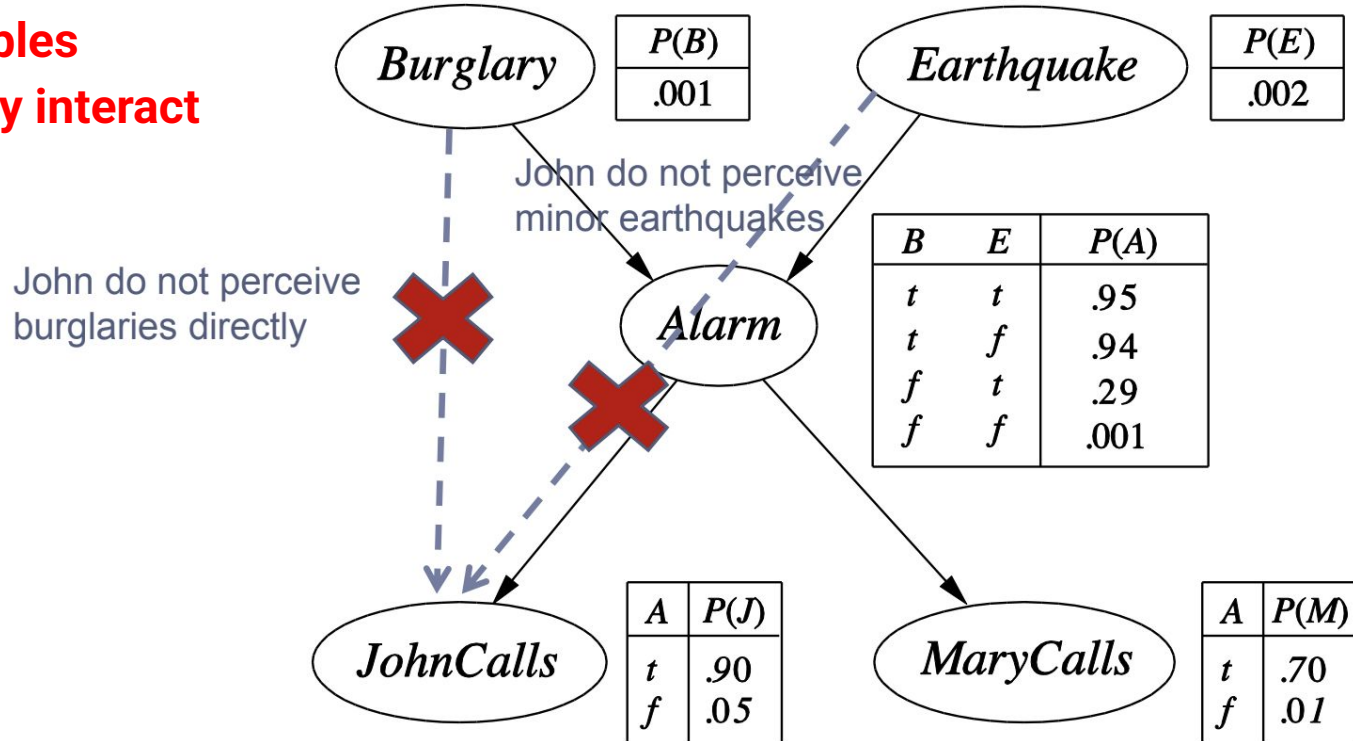


Burglary Example's BN:



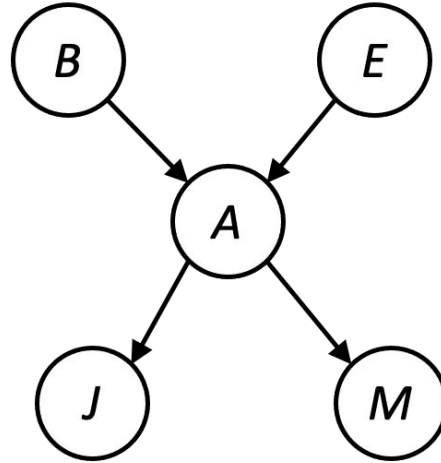
Burglary Example's BN:

describe how
variables
locally interact



Joint Distribution from BN in Burglary Example

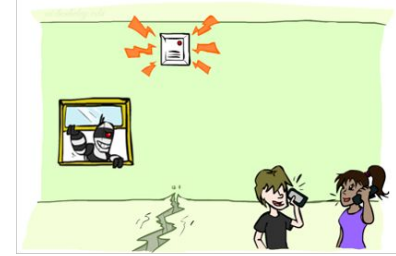
B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

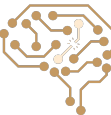
A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99



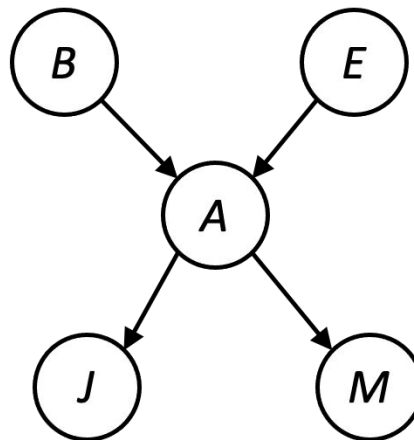
B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$P(+b, -e, +a, -j, +m) =$$



Joint Distribution from BN in Burglary Example (cont.)

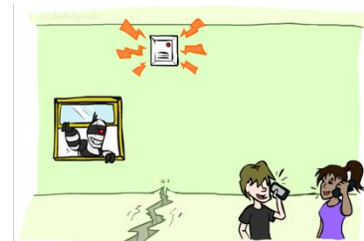
B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

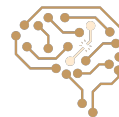
A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99



B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$\begin{aligned}
 P(+b, -e, +a, -j, +m) &= \\
 P(+b)P(-e)P(+a|+b, -e)P(-j|+a)P(+m|+a) &= \\
 0.001 \times 0.998 \times 0.94 \times 0.1 \times 0.7 &
 \end{aligned}$$



Size of a Bayes Net

- How big is a joint distribution over N Boolean variables?

$$2^N$$

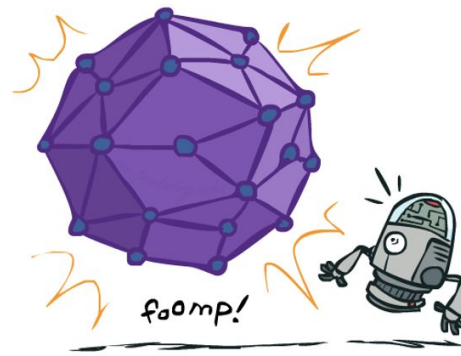
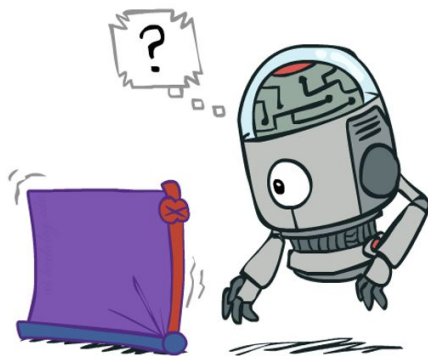
- How big is an N -node net if nodes have up to k parents?

$$O(N * 2^{k+1})$$

- Both give you the power to calculate

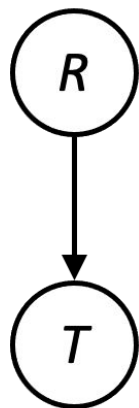
$$P(X_1, X_2, \dots, X_n)$$

- BNs: Huge space savings!
- Also easier to elicit local CPTs
- Also faster to answer queries (coming)



Example: Traffic

- Causal direction



$P(R)$

+r	1/4
-r	3/4

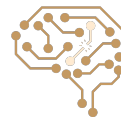
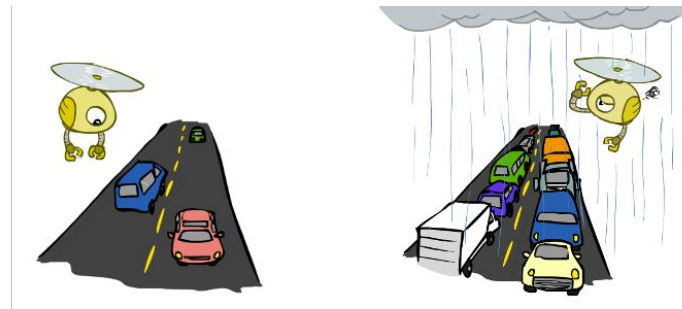
$P(T|R)$

+r	+t	3/4
	-t	1/4

-r	+t	1/2
	-t	1/2

$P(T, R)$

+r	+t	3/16
+r	-t	1/16
-r	+t	6/16
-r	-t	6/16



Models

Models describe how (a portion of) the world works

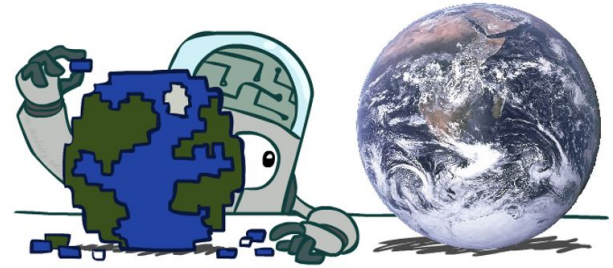
Models are always simplifications

May not account for every variable

May not account for all interactions between variables

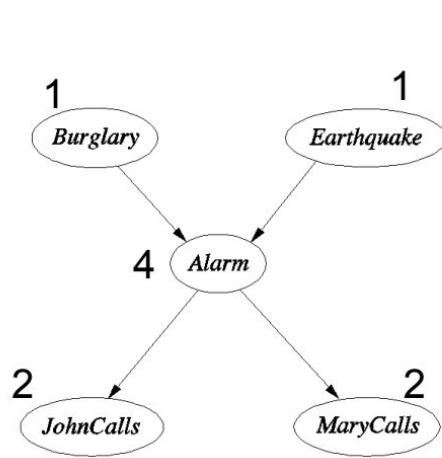
“All models are wrong; but some are useful.”

– George E. P. Box

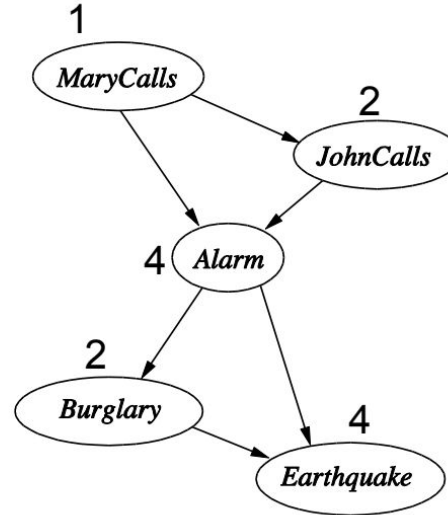


Burglary Example

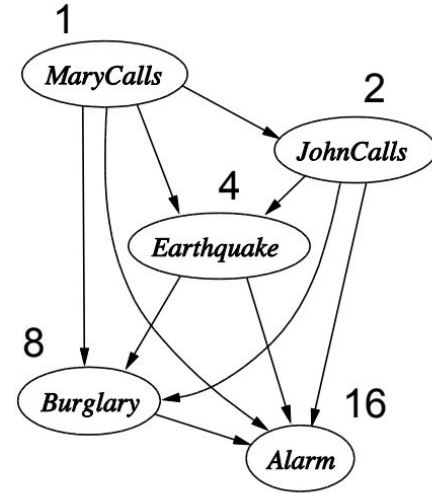
The structure of the network and so the number of required probabilities for different node orderings can be different



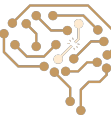
$$1 + 1 + 4 + 2 + 2 = 10$$



$$1 + 2 + 4 + 2 + 4 = 13$$



$$1 + 2 + 4 + 8 + 16 = 31$$



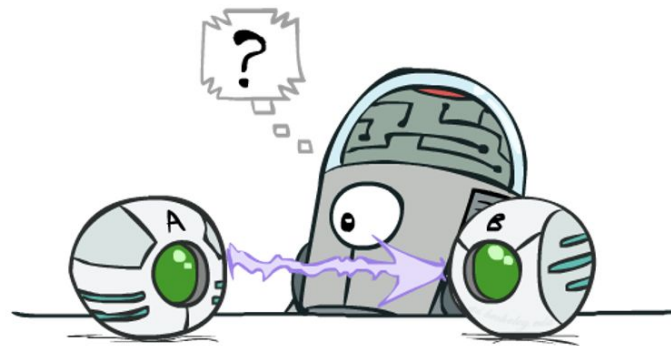
Causality?

When Bayes' nets reflect the true causal patterns:

- Often simpler (nodes have fewer parents)
- Often easier to think about
- Often easier to elicit from experts

BNs need not actually be causal

- Sometimes no causal net exists over the domain (especially if variables are missing)
- E.g. consider the variables *Traffic* and *Drips*
- End up with arrows that reflect correlation, not causation



What do the arrows really mean?

- Topology may happen to encode causal structure
- **Topology really encodes conditional independence**

$$P(x_i | x_1, \dots, x_{i-1}) = P(x_i | \text{parents}(X_i))$$



Constructing Bayesian Network

Nodes:

determine the set of variables and order them as $X_1, \dots, X_n$

(More compact network if causes precede effects)

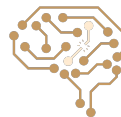
Links:

for $i = 1$ to n

- 1) select a minimal set of parents for X_i from $X_1, \dots, X_{i-1}$ such that

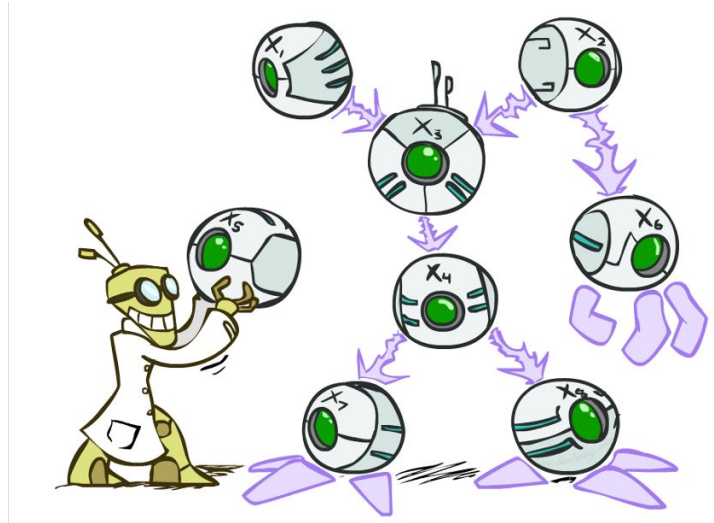
$$\mathbf{P}(X_i \mid \text{Parents}(X_i)) = \mathbf{P}(X_i \mid X_1, \dots, X_{i-1})$$

- 2) For each parent insert a link from the parent to X_i
- 3) CPT creation based on $\mathbf{P}(X_i \mid X_1, \dots, X_{i-1})$

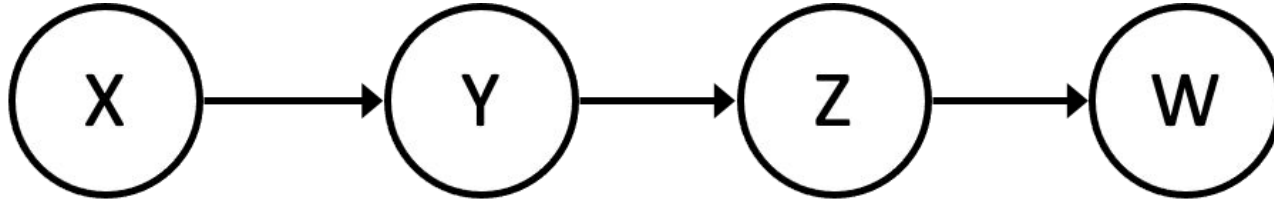


Bayes' Nets

- So far: how a Bayes' net encodes a joint distribution
- Next: how to answer queries about that distribution



Example: Conditional Independence in BNs



- Is Z independent of X , given Y ?
- Is W independent of Y , given Z ?
- Is W independent of X , given Z ?
- Is W independent of X , given Y ?



Next Session: Bayes Nets Independence

